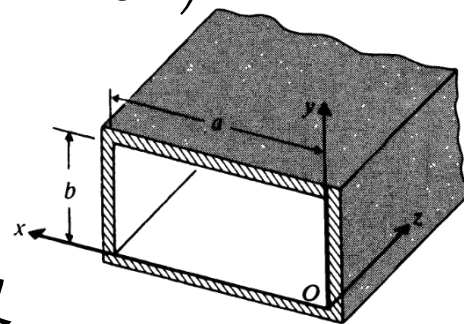


$$E_y = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right)$$

$$H_x = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right)$$

$$E_x = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right)$$

$$H_y = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right)$$



- 1) 推导波导色散关系 (β 与频率 f 及截止频率 f_c 关系)
- 2) 推导波导中三个波长关系, 并说明三个波长的物理意义
- 3) 推导rectangular waveguide中TE和TM模式的电磁场各个分量及色散关系
- 4) **P.10-16** Calculate and list in ascending order the cutoff frequencies (in terms of the cutoff frequency of the dominant mode) of an $a \times b$ rectangular waveguide for the following modes: TE_{01} , TE_{10} , TE_{11} , TE_{02} , TE_{20} , TM_{11} , TM_{12} , and TM_{22} (a) if $a = 2b$ and (b) if $a = b$.

P.10-16 $(f_c)_{mn} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{1}{2a\sqrt{\mu\epsilon}} F(m,n).$

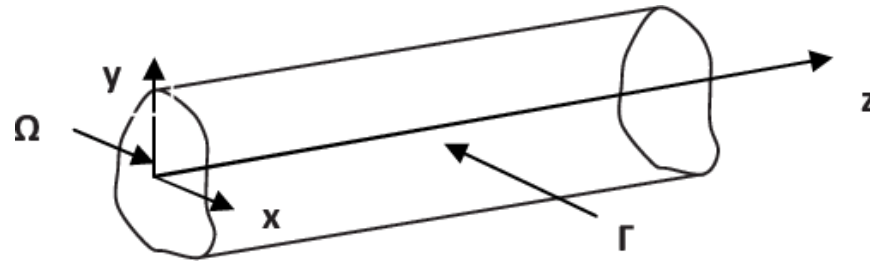
a) $a=2b$, $F(m,n) = \sqrt{m^2 + 4n^2}$

Modes	$F(m,n)$
TE_{10}	1
TE_{01}, TE_{20}	2
TE_{11}, TM_{11}	$\sqrt{5}$
TE_{02}	4
TM_{12}	$\sqrt{17}$
TM_{21}	$\sqrt{20}$

b) $a=b$, $F(m,n) = \sqrt{m^2 + n^2}$

Modes	$F(m,n)$
TE_{10}, TE_{01}	1
TE_{11}, TM_{11}	$\sqrt{2}$
TE_{02}, TE_{20}	2
TM_{12}	$\sqrt{5}$
TM_{21}	$2\sqrt{2}$

Helmholtz's eqs



$$e^{-\gamma z} e^{j\omega t}$$

$$\gamma = \alpha + j\beta$$

$$\nabla^2 E + k^2 E = 0$$

$$k^2 = \omega^2 \mu \epsilon$$

k – wavenumber

$$\frac{\partial^2}{\partial x^2} E + k_x^2 E = 0$$

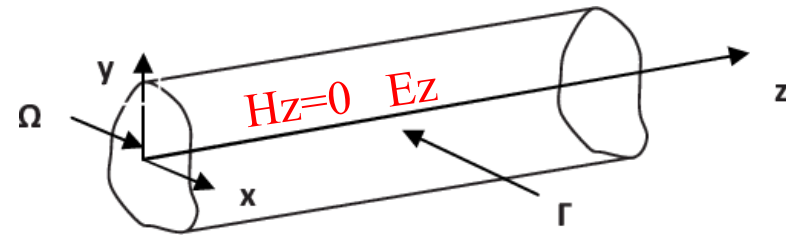
$$\frac{\partial^2}{\partial y^2} E + k_y^2 E = 0$$

$$\frac{\partial^2}{\partial z^2} E + k_z^2 E = 0$$

TM waves

$$\nabla_T^2 E + h^2 E = 0$$

$$\nabla_T^2 E_z + h^2 E_z = 0$$



$$h^2 = k^2 + \gamma^2 = k^2 - \beta^2$$

- $H_z=0$

$$H_x = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right)$$

$$H_y = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right)$$

$$E_x = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right)$$

$$E_y = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right)$$



$$H_x^0 = \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial y},$$

$$H_y^0 = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial x},$$

$$E_x^0 = -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial x},$$

$$E_y^0 = -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial y}.$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ E_x & E_y & E_z \end{vmatrix}$$

$$e^{-\gamma z} e^{j\omega t}$$

$$= -j\omega\mu \{ \hat{x}H_x + \hat{y}H_y + \hat{z}H_z \}$$

TE waves

$E_z=0$ H_z

$$\nabla_T^2 H + h^2 H = 0$$

$$\nabla_T^2 H_z + h^2 H_z = 0$$

$$\begin{aligned} h^2 &= k^2 + \gamma^2 \\ &= k^2 - \beta^2 \end{aligned}$$


$$\begin{aligned} H_x &= -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial \mathbf{E}_z}{\partial y} \right) & H_x^0 &= -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial x}, \\ H_y &= -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial \mathbf{E}_z}{\partial x} \right) & H_y^0 &= -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial y}, \\ E_x &= -\frac{1}{h^2} \left(\gamma \frac{\partial \mathbf{E}_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right) & E_x^0 &= -\frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial y}, \\ E_y &= -\frac{1}{h^2} \left(\gamma \frac{\partial \mathbf{E}_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right) & E_y^0 &= \frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial x}. \end{aligned}$$

TM wave along Parallel plate WG

$$E_z(y) = A_n \sin\left(\frac{n\pi}{b} y\right)$$

$$H_x(y) = \frac{j\omega\epsilon}{h} A_n \cos\left(\frac{n\pi}{b} y\right)$$

$$E_y(y) = -\frac{\gamma}{h} A_n \cos\left(\frac{n\pi}{b} y\right)$$

Propagation constant

$$\gamma = \sqrt{h^2 - k^2} = \sqrt{\left(\frac{n\pi}{b}\right)^2 - \omega^2 \mu \epsilon}$$

$$k^2 = \left(\frac{n\pi}{b}\right)^2 + \beta^2 = h^2 + \beta^2$$

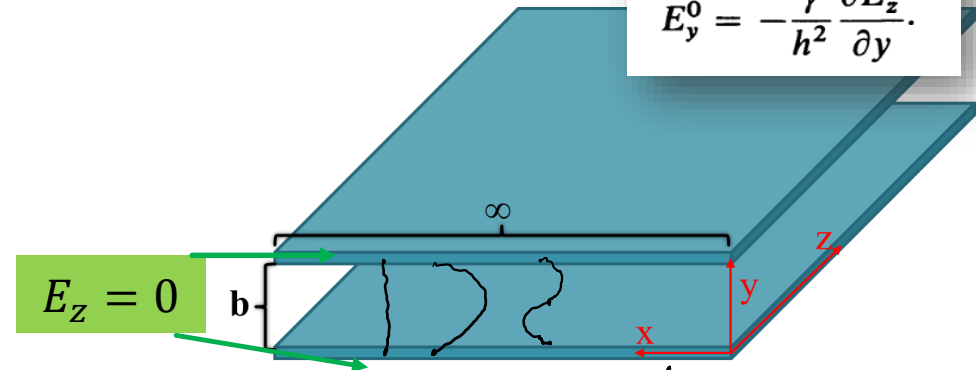
$$h = \frac{n\pi}{b}$$

Cutoff frequency

$$\gamma = 0 \rightarrow f_c = \frac{n}{2b\sqrt{\mu\epsilon}}$$

$$\lambda_c = \frac{2b}{n}$$

$$\begin{aligned} H_x^0 &= \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial y}, \\ H_y^0 &= -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial x}, \\ E_x^0 &= -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial x}, \\ E_y^0 &= -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial y}. \end{aligned}$$



$$\begin{aligned} \lambda_y &= \infty & \lambda_y &= 2b & \lambda_y &= b \\ h &= K_y < 0 & h &= \frac{\pi}{b} & h &= \frac{\pi}{b} \end{aligned}$$

$$\gamma = j\beta$$

Dominant mode: TM0 mode is TEM mode

TE wave along Parallel plate WG

$$H_z(y) = B_n \cos\left(\frac{n\pi}{b} y\right)$$

$$E_x(y) = \frac{j\omega\mu}{h} B_n \sin\left(\frac{n\pi}{b} y\right)$$

$$H_y(y) = \frac{\gamma}{h} B_n \sin\left(\frac{n\pi}{b} y\right)$$

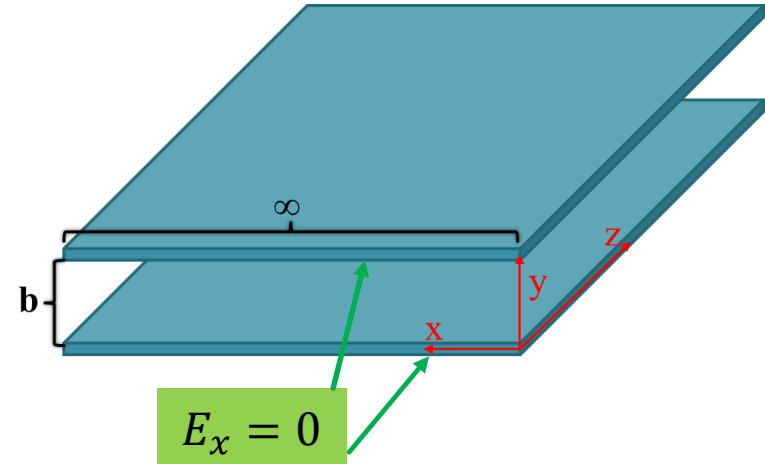
Propagation constant

$$\gamma = \sqrt{h^2 - k^2} = \sqrt{\left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\epsilon}$$

$$k^2 = \left(\frac{n\pi}{b}\right)^2 + \beta^2 = h^2 + \beta^2$$

Cutoff frequency

$$\gamma = 0 \rightarrow f_c = \frac{n}{2b\sqrt{\mu\epsilon}}$$



$$\begin{aligned} H_x^0 &= -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial x}, \\ H_y^0 &= -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial y}, \\ E_x^0 &= -\frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial y}, \\ E_y^0 &= \frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial x}. \end{aligned}$$

TM modes in Rec. WG

$H_z=0$

$$E_z(x, y) = E_0 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

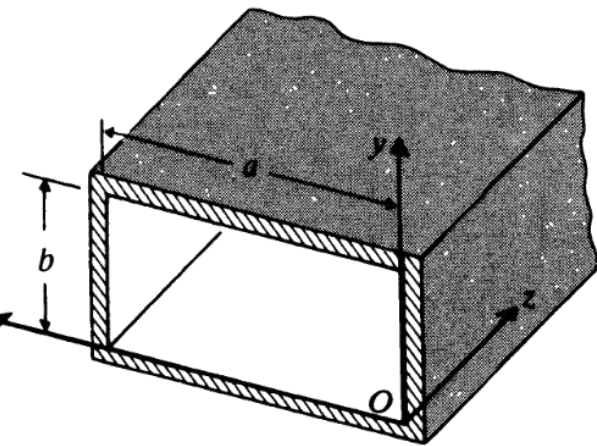
$$E_x(x, y) = -\frac{\gamma}{h^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$E_y(x, y) = -\frac{\gamma}{h^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$H_x(x, y) = \frac{j\omega\epsilon}{h^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$H_y(x, y) = -\frac{j\omega\epsilon}{h^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$\beta = \sqrt{\omega^2\mu\epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$



$$H_x^0 = \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial y},$$

$$H_y^0 = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial x},$$

$$E_x^0 = -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial x},$$

$$E_y^0 = -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial y}.$$

- For TM modes in Rect WG, neither m nor n can be zero.

TE modes in Rec. WG

$$H_z(x, y) = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

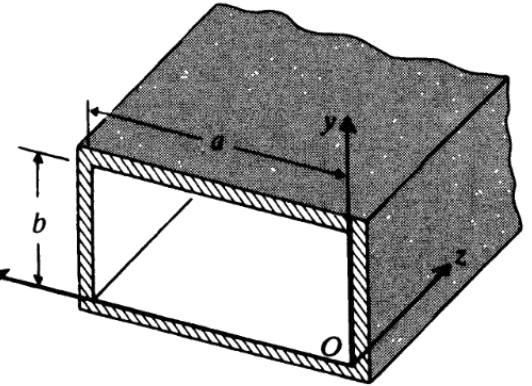
$$H_x(x, y) = \frac{\gamma}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$H_y(x, y) = \frac{\gamma}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$E_x(x, y) = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$E_y(x, y) = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

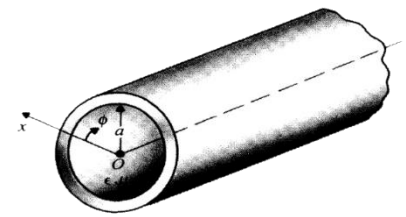
$$\beta = \sqrt{\omega^2\mu\epsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$



$$\begin{aligned} H_x^0 &= -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial x}, \\ H_y^0 &= -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial y}, \\ E_x^0 &= -\frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial y}, \\ E_y^0 &= \frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial x}. \end{aligned}$$

- For TE modes in Rect WG, either m nor n (not both) can be zero.

Discussion



- Single-conductor waveguides cannot support TEM waves
 - Magnetic flux lines always close upon themselves
 - **Ampere circuital law: Line integral** of the magnetic field H around any **closed loop** in a transverse plane must **equal** the **sum of the longitudinal conduction and displacement currents through the loop**.

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s}. \quad (7-54b)$$

Without the inner conductor $\sigma=0$,
no longitudinal conduction current J .

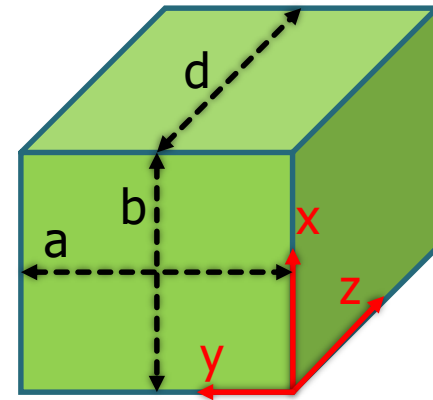
TEM waves does not have z components $E_z=0$,
no longitudinal displacement current $\partial \mathbf{D} / \partial t$

- Thus, **$H=0$** , then **$E=0$**

10-7

Cavity Resonators

Rect. Cavity resonators



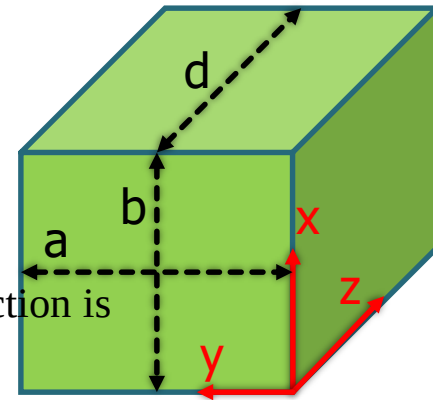
Why do we want to use resonators

- To provide frequency selection/generation functions -- filtering

A rectangular waveguide with both ends closed by a conducting wall

- The designation of TM and TE modes in a resonator is not unique, since we are free to choose x or y or z as the direction of propagation. For example, a TE mode with respect to the z axis could be a TM mode with respect to the y axis.
- When the cavity or resonator is closed, no wave is propagating. Wave inside the resonator will be reflected by the walls for multiple times and hence setting up standing waves.
- Since there are 3 pairs of walls forming restriction on the EM wave, we will need 3 symbol subscript to designate a TM or TE standing wave pattern. TM_{mnp} and TE_{mnp}

TM_{mnp} modes



Field components of a resonator

- In rectangular WG, the longitudinal variation for a wave travelling in the +z direction is described by $e^{-j\beta z}$
- For standing wave case, superposition of two waves, $e^{-j\beta z}$ and $e^{j\beta z}$ with same amplitude will give a standing wave of the form $\sin(\beta z)$ or $\cos(\beta z)$.
- For tangential 切向 components of E field in the xy plane, E_x, E_y must be $\sin()$ form to make them be 0 at $z=0$ and $z=d$. Other components. Recall, γ comes from $\partial/\partial z$, then the other components will be $\cos()$ form.

$$\begin{aligned} H_x^0 &= \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial y}, \\ H_y^0 &= -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial x}, \\ E_x^0 &= -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial x}, \\ E_y^0 &= -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial y}. \end{aligned}$$

$$E_z(x, y) = E_0 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$E_x(x, y) = -\frac{\gamma}{h^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$E_y(x, y) = -\frac{\gamma}{h^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$H_x(x, y) = \frac{j\omega\epsilon}{h^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$H_y(x, y) = -\frac{j\omega\epsilon}{h^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$\bullet E_z(x, y) = E_0 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$$

$$\bullet E_x(x, y) = -\frac{1}{h^2} \left(\frac{m\pi}{a}\right) \left(\frac{p\pi}{d}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$$

$$\bullet E_y(x, y) = -\frac{1}{h^2} \left(\frac{n\pi}{b}\right) \left(\frac{p\pi}{d}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$$

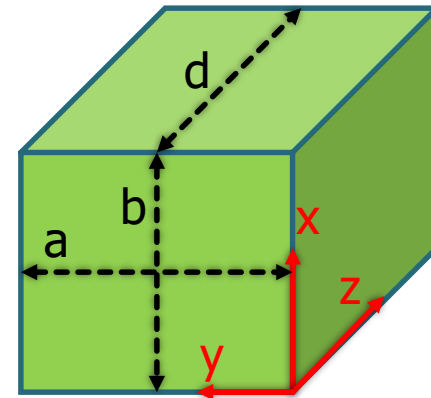
$$\bullet H_x(x, y) = \frac{j\omega\epsilon}{h^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$$

$$\bullet H_y(x, y) = -\frac{j\omega\epsilon}{h^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$$

TE_{mnp} modes

Field components of a resonator

- In rectangular WG, the longitudinal variation for a wave travelling in the +z direction is described by $e^{-j\beta z}$
- For standing wave case, superposition of two waves, $e^{-j\beta z}$ and $e^{j\beta z}$ with same amplitude will give a standing wave of the form $\sin(\beta z)$ or $\cos(\beta z)$
- For tangential components of E field in the xy plane, E_x , E_y must be $\sin()$ form to make them be 0 at $z=0$ and $z=d$. Other components. Recall, γ comes from $\partial/\partial z$, then the other components will be $\cos()$ form.



$$H_z(x, y) = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$\bullet H_z(x, y) = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$$

$$H_x(x, y) = \frac{\gamma}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$\bullet H_x(x, y) = -\frac{1}{h^2} \left(\frac{m\pi}{a}\right) \left(\frac{p\pi}{d}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$$

$$H_y(x, y) = \frac{\gamma}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$\bullet H_y(x, y) = -\frac{1}{h^2} \left(\frac{n\pi}{b}\right) \left(\frac{p\pi}{d}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$$

$$E_x(x, y) = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right)$$

$$\bullet E_x(x, y) = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$$

$$E_y(x, y) = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right)$$

$$\bullet E_y(x, y) = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$$

$$\omega_{mnp} = \frac{1}{\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$$

or

$$f_{mnp} = \frac{u}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2} \quad (\text{Hz}).$$

degenerate(简并) **modes**: different modes having the same resonant frequency
Thus, TM_{mnp} and TE_{mnp} modes are always degenerate if none of the mode indices is zero.

dominant mode: the mode with the lowest resonant frequency.

For TM_{mnp} modes, neither m nor n can be zero, but p can be zero;

For TE_{mnp} modes, either m or n (not both) can be zero, but p cannot be zero.

Thus, the modes of the lowest orders are **TM₁₁₀, TE₀₁₁, TE₁₀₁**.

(D. K. Cheng, p. 586)

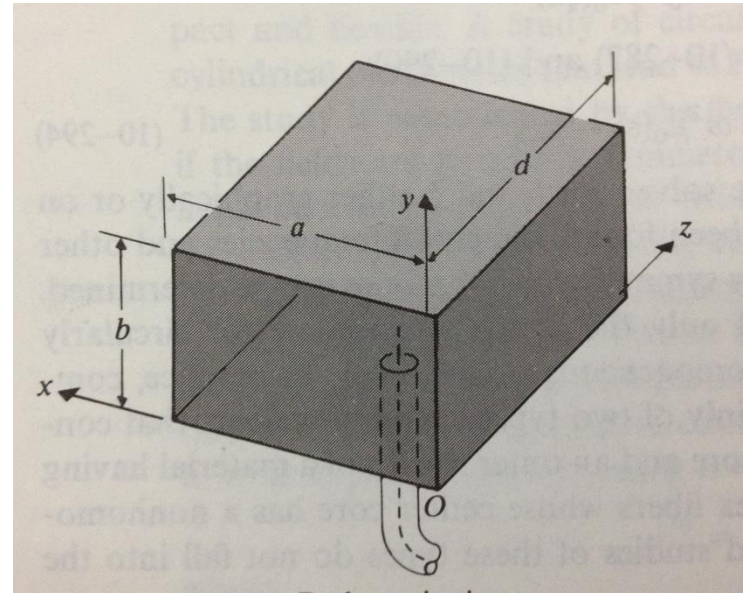
Cavity resonators

z-direction: propagation direction

$$f_{mnp} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

***TM: Neither m nor n can be zero
 p can be zero***

***TE: Either m or n can be zero (not both)
 p can not be zero***



The lowest mode TM_{110}

The lowest mode TE_{101} or TE_{011}

10-15 Determine the dominant modes and their frequencies in an air-filled rectangular cavity resonator for (a) $a > b > d$, (b) $a > d > b$, and (c) $a = b = d$, where a , b , and d are the dimensions in the x -, y -, and z -directions, respectively.

the modes of the lowest orders are

$$TM_{110}, \quad TE_{011}, \quad TE_{101}.$$

(a) For $a > b > d$: the lowest resonant frequency is

$$f_{110} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}.$$

TM_{110} is the dominant mode.

(b) For $a > d > b$: the lowest resonant frequency is

$$f_{101} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{d^2}}.$$

TE_{101} is the dominant mode.

(c) For $a = d = b$, all three of the lowest-order modes (namely, TM_{110} , TE_{011} , and TE_{101}) have the same field patterns. The resonant frequency of these degenerate modes is

$$f_{110} = \frac{c}{\sqrt{2}a}.$$

- $\mathbf{E}_z(x, y) = E_0 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$
- $E_x(x, y) = -\frac{1}{h^2} \left(\frac{m\pi}{a}\right) \left(\frac{p\pi}{d}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$
- $E_y(x, y) = -\frac{1}{h^2} \left(\frac{n\pi}{b}\right) \left(\frac{p\pi}{d}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$
- $\mathbf{H}_x(x, y) = \frac{j\omega\epsilon}{h^2} \left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$
- $\mathbf{H}_y(x, y) = -\frac{j\omega\epsilon}{h^2} \left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$

TM_{110} , TE_{011} , and TE_{101}

- | | |
|--|--|
| • $\mathbf{H}_z(x, y) = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$ | • $\mathbf{H}_z(x, y) = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$ |
| • $H_x(x, y) = -\frac{1}{h^2} \left(\frac{m\pi}{a}\right) \left(\frac{p\pi}{d}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$ | • $\mathbf{H}_x(x, y) = -\frac{1}{h^2} \left(\frac{m\pi}{a}\right) \left(\frac{p\pi}{d}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$ |
| • $\mathbf{H}_y(x, y) = -\frac{1}{h^2} \left(\frac{n\pi}{b}\right) \left(\frac{p\pi}{d}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$ | • $H_y(x, y) = -\frac{1}{h^2} \left(\frac{n\pi}{b}\right) \left(\frac{p\pi}{d}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \cos\left(\frac{p\pi}{d}z\right)$ |
| • $\mathbf{E}_x(x, y) = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$ | • $E_x(x, y) = \frac{j\omega\mu}{h^2} \left(\frac{n\pi}{b}\right) H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$ |
| • $E_y(x, y) = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$ | • $\mathbf{E}_y(x, y) = -\frac{j\omega\mu}{h^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{p\pi}{d}z\right)$ |

Quality factor (Q factor 品质因子)

A cavity resonator stores energy in the electric and magnetic fields for any particular mode pattern. In any practical cavity the walls have a finite conductivity; that is, a nonzero surface resistance, and the resulting power loss causes a decay of the stored energy. The **quality factor**, or Q , of a resonator, like that of any resonant circuit, is a measure of the bandwidth of the resonator and is defined as

$$Q = 2\pi \frac{\text{Time-average energy stored at a resonant frequency}}{\text{Energy dissipated in one period of this frequency}}. \quad (10-313)$$

(Dimensionless)

Let W be the total time-average energy in a cavity resonator. We write

$$W = W_e + W_m, \quad (10-314)$$

where W_e and W_m denote the energies stored in the electric and magnetic fields, respectively. If P_L is the time-average power dissipated in the cavity, then the energy dissipated in one period is P_L divided by frequency, and Eq. (10-313) can be written as

$$Q = \frac{\omega W}{P_L} \quad (\text{Dimensionless})$$

Life time and Q factor (品质因子)

For a resonator, one always need to excite it by input energy, or bring the energy out of the resonator. Therefore, there is always a dissipation of energy in the resonator. The quality factor **Q** is a universal measurement.

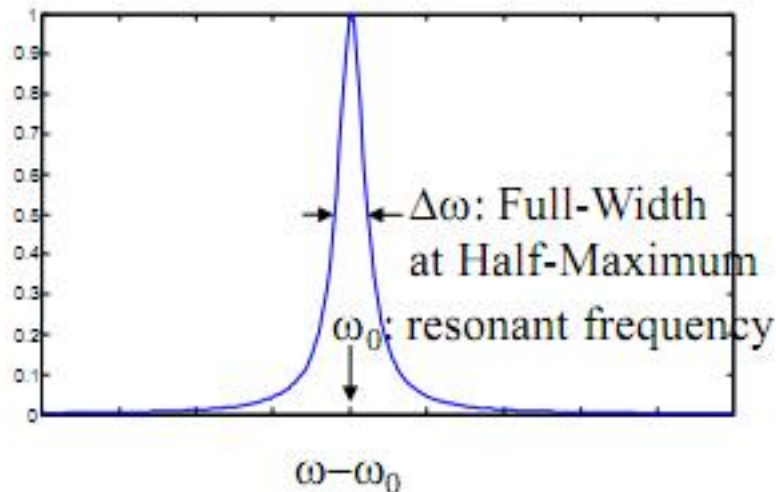
$$Q = \omega_0 \times (\text{Energy stored } U) / (\text{Power dissipated } -dU/dt)$$

$$Q = \omega_0 \frac{U}{P}$$

Then we have

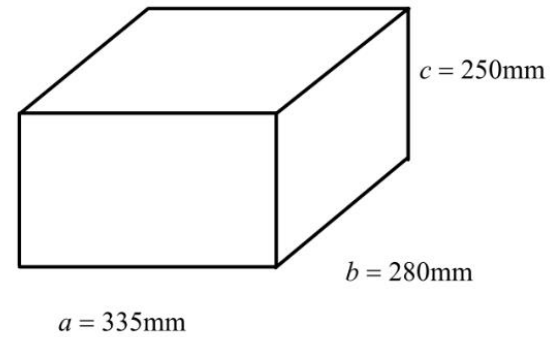
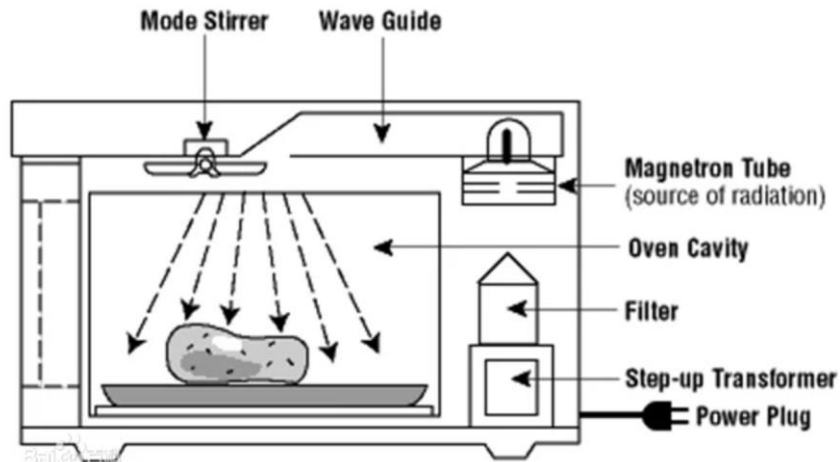
$$U(t) = U_0 e^{-\omega_0 t / Q}$$

Definition: Life time a photon in a resonator is $\tau_p = Q / \omega_0$



The frequency spectrum of a resonator has a finite width around the resonant frequency

$$Q = \omega_0 / \Delta\omega.$$



(2450 ± 10) MHz

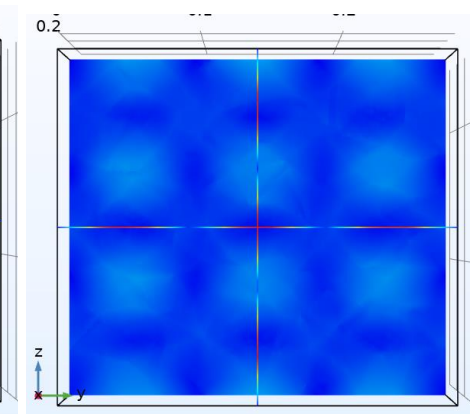
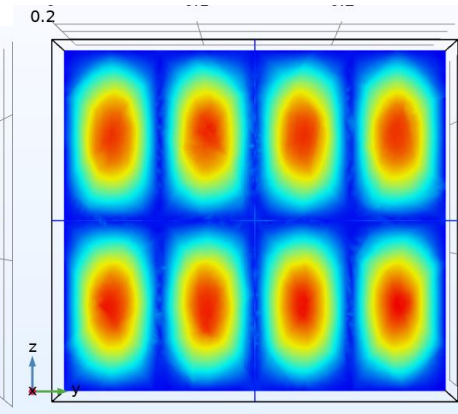
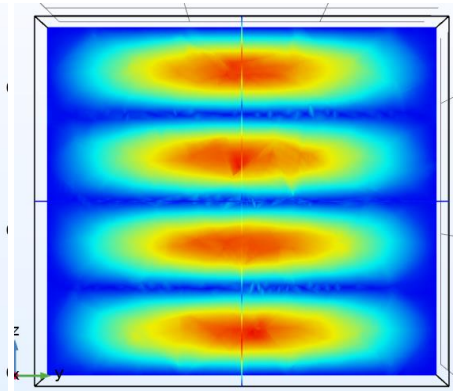
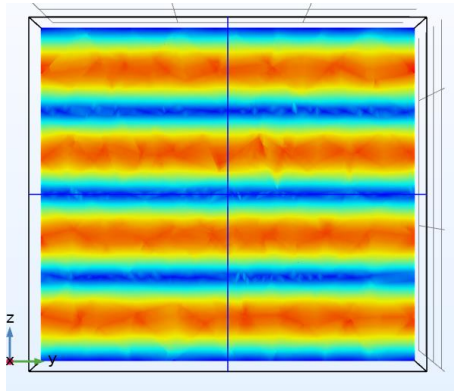
m	n	p
0	1	4
0	4	2
1	3	3
1	0	4
1	1	4

(2450 ± 50) MHz

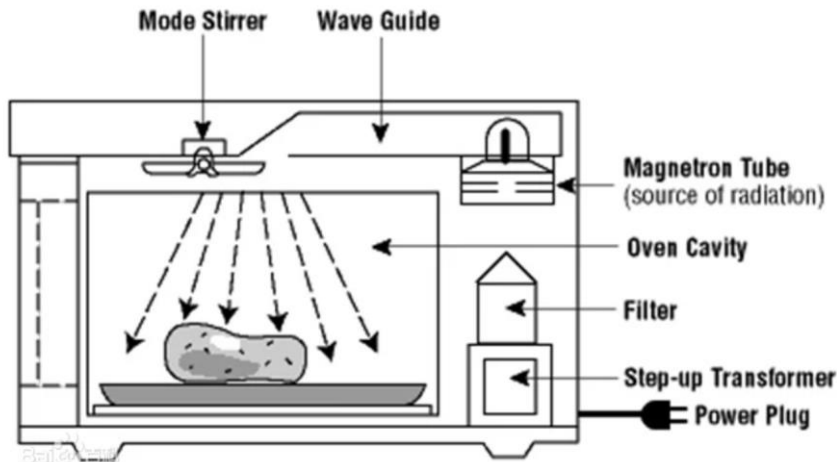
m	n	p
1	1	4
1	4	2
5	2	0
4	3	0
0	3	3
3	3	2
4	2	2

$$f_{mnp} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

Bisection of yz plane



均匀加热



- 引入更多模式使得电场分布更均匀
- 旋转加热样品（托盘式微波炉）
- 旋转电场（平板式微波炉）

通过旋转金属搅拌架，通过边界条件的周期性变化实现

变频微波炉需要改变谐振腔参数么？

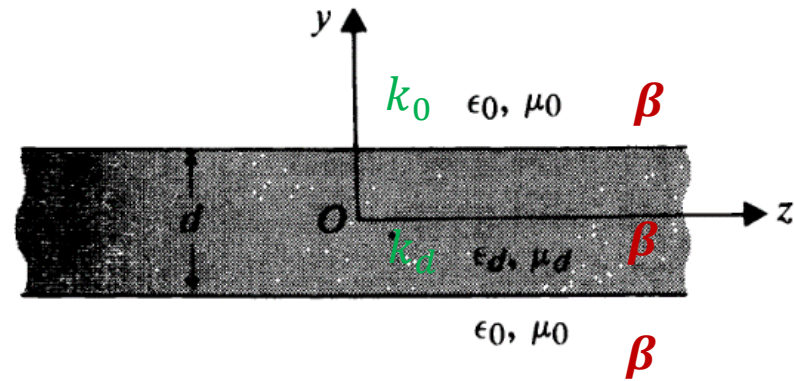
变频 $\neq$ 改变频率

一般微波炉的变频指的是改变占空比，从而调制输出功率的时间序列，改变占空比从而实现节能。

10-6

Dielectric Waveguides

TM modes in a Slab WG



$H_z=0$, no variance in the x direction,

- $E_z(x, y, z) = E_z(y)e^{-\gamma z} = E_z(y)e^{-j\beta z}$

$$\nabla^2 E + k^2 E = 0 \quad \nabla_T^2 E_z + h^2 E_z = 0 \rightarrow \left(\frac{\partial^2}{\partial y^2} + h^2 \right) E_z(y) = 0$$

$$\nabla_T^2 E + h^2 E = 0$$

For $|y| < \frac{d}{2}$ $E_z(y) = E_o \sin(k_y y) + E_e \cos(k_y y)$

- $k_y^2 \equiv k_d^2 - \beta^2$, dispersion relation

For $|y| > \frac{d}{2}$, waves must be exponentially decaying, so that there is no leak.

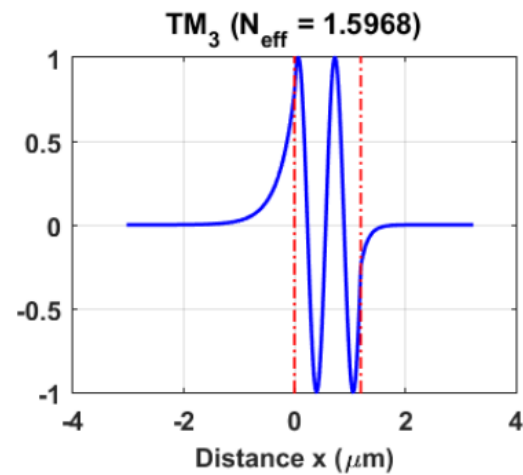
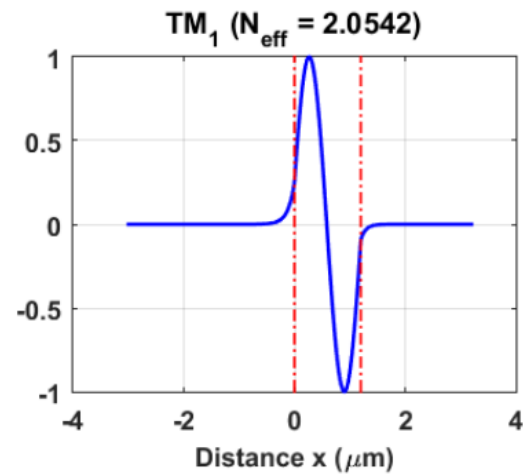
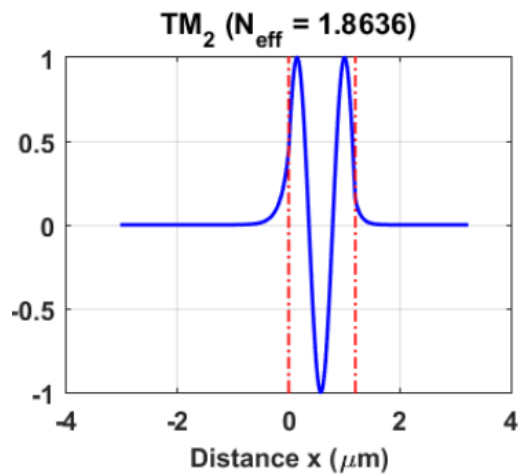
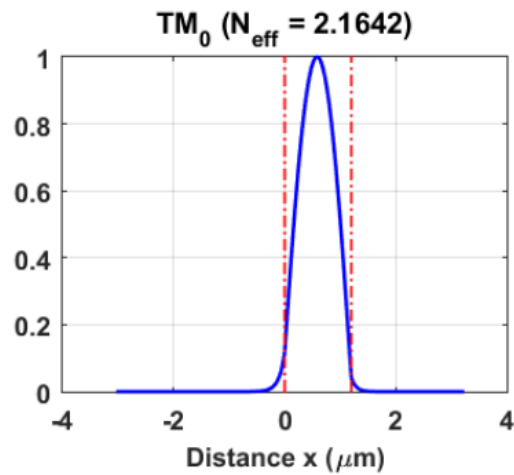
Same β

$$E_z(y) = \begin{cases} c_u e^{-\alpha(y-\frac{d}{2})}, & y > \frac{d}{2} \\ c_l e^{\alpha(y+\frac{d}{2})}, & y < -\frac{d}{2} \end{cases}$$

- $(j\alpha)^2 = k_0^2 - \beta^2$, dispersion relation,

$$\alpha^2 + k_y^2 = k_d^2 - k_0^2$$

H_x



E_z

E_y

H_x